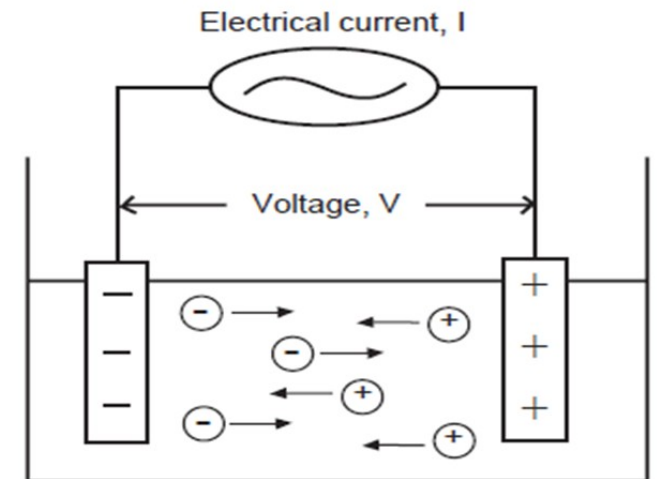


Conductometry

Electrolytic conductance

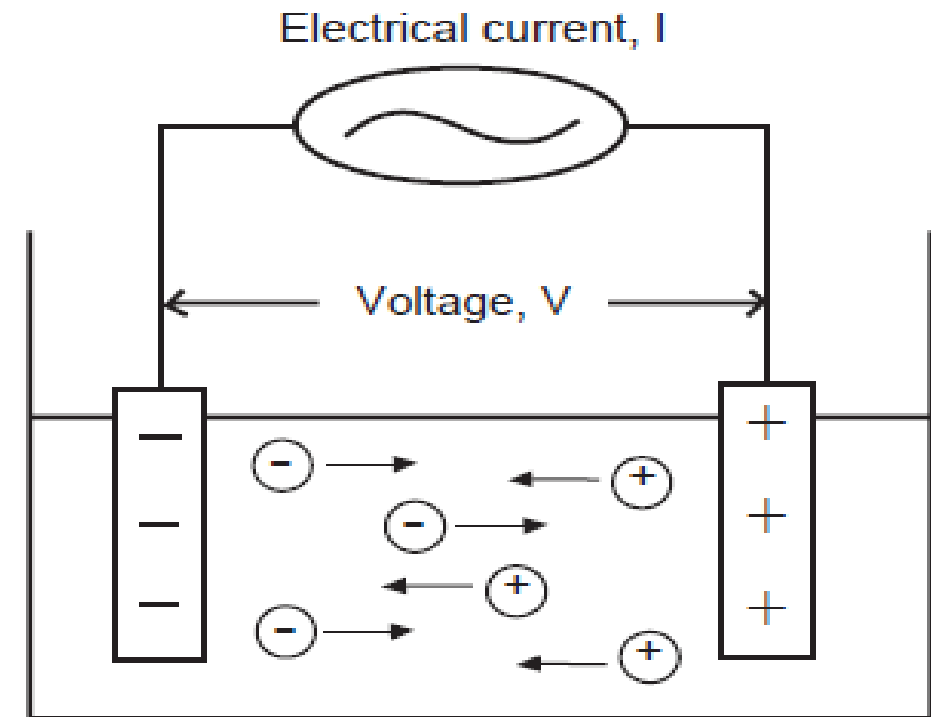
- **Electric current** passes through **metallic conductors**, e.g. wires of iron or copper in the **form of electron flow**.
- But conduction of electricity through an **electrolyte solution** involves migration of **positively charged species towards the cathode** and **negatively charged species towards the anode**, i.e. **current is carried by ions present in solution**



Migration of ions in solution

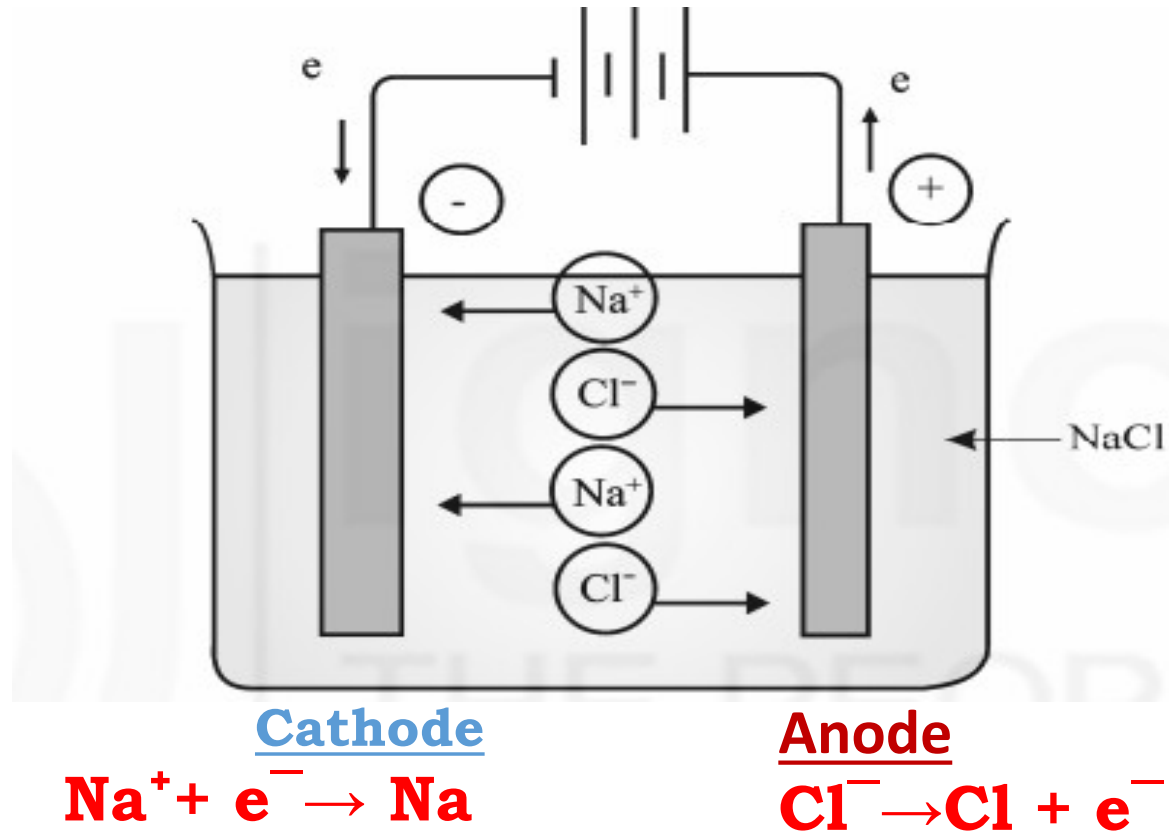
- **Conductometry** means measuring the **conductivity of ionic solutions** caused by the **mobility of ions towards respective electrode** in the presence of **an electric potential**.

- **Conductivity is measured by Conductometer**



- **Electrolytic conduction, in which charges carried by ions, will not occur unless the ions of the electrolyte are free to move.**
- **Electrolytic conduction** is exhibited principally by **molten salts and by aqueous solutions of electrolytes**
- Conductometry means **measuring the conductivity of an electrolyte or molten salts.**
- a **conductometer** measures the electrical conductivity of **ionic solutions.**

- The principle of electrolytic conduction is best illustrated by reference to an electrolytic cell such as that shown in



Electrolysis of molten sodium chloride

- The electrons are received from the **negative end of the external battery** by the negative electrode of the cell.
- These are used up in the **reduction reaction** at this electrode.
- The numbers of **electrons received at the negative electrode** are **given back to the positive end of the external** battery from the positive electrode of the cell where electrons are released as the result of **oxidation reaction**.
 - ✓ *Within the cell, current is carried by the movement of ions; cations moves towards negative electrode called the cathode and anions towards the positive electrode called anode.*
 - ✓ *This movement of ions give rise to what is known as the **electrolytic conduction***

- Electrolytic conduction depends on the **mobility of ions** and **anything that inhibits the motion of ions** causes **resistance to current flow**.
- Factors that influence the electrical conductivity of solutions of electrolytes include **interionic attraction**, **solvation of ions**, and **viscosity of solvents etc...**
- The **average KE of the solute ions increases** as the **temperature is raised** and, therefore, the **resistance of electrolytic conductors generally decreases**.
 - ✓ That is, conduction increases as the temperature is raised.

Electrolytes and non-electrolytes

- The ionic compounds, **which furnish ions in solution and conduct electric current, are electrolytes e.g. NaCl, KCl, etc.**
- There are **covalent compounds**, which also conduct electric current in solutions.

These include HCl, CH₃COOH, etc.

- All other substances **which do not produce ions in solutions are called non-electrolytes**

e.g. sugar cane, benzene, carbontetrachloride, etc.

Strong and weak electrolytes

- Electrolytes can be classified as **strong or weak**.
 - ✓ related to their **extent of ionization not on concentration**.
- ***Strong electrolytes***: completely ionized in aqueous solutions
e.g. **NaCl, NH₄Cl, KNO₃, HCl, HBr**, etc.
- ***Weak electrolytes***: ionize only to a certain extent
e.g. **CH₃COOH, HCN**, etc.

Strong and weak electrolytes...

- The behavior of electrolytes also depends on **the nature of solvents**,

e.g.

➤ NaCl behaves as **strong electrolyte** whereas acetic acid as **a weak electrolyte in water**.

➤ On the other hand, when dissolved in **ammonia** both NaCl and acetic acid show **comparable behavior towards electricity**.

Conductance of solutions

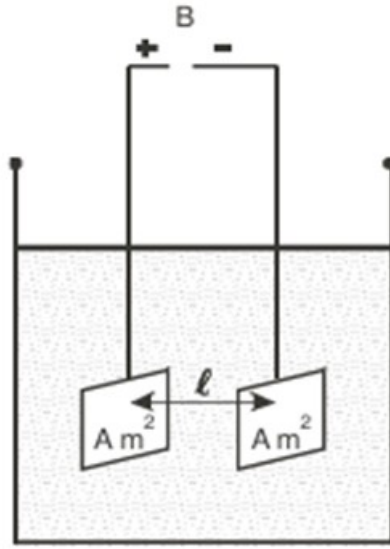
- The **ease of flow of electric current** through a body is called its **conductance**.
- In **metallic conductors** it is caused by the **movement of electrons**, while in **electrolytic solutions** it is caused by **ions of electrolyte**.
- The **electrolytic conductance "G"** of a medium is equal to the reciprocal of its **electrical resistance "R "in ohms**.

$$G = \frac{1}{R}$$

- Ohm's Law states that the current I (amperes) is directly proportional to the applied electromotive force E (volts) and inversely proportional to the resistance, R (ohms) of the conductor:

$$I = \frac{E}{R}, \quad E = IR$$

- Since a solution is **a three-dimensional conductor**, the resistance will depend on the spacing (l) and area (A) of the electrodes



- Consider the electrolytic cell shown above, its two electrodes are having a cross sectional area of **$A \text{ m}^2$** and separated by **$l \text{ m}$** .
- The resistance (R) of the electrolyte solution present between the two electrodes is:

- **$R \sim l$ and $R \sim \frac{1}{A}$**

$$\therefore R = \rho \frac{l}{A}$$

Where:
 ρ =resistivity

- In SI units, l and A are measured in meters and square meters respectively, and the resistance is expressed in ohm, Ω (omega).
- Therefore, the unit of ρ is ohm meters ($\Omega \text{ m}$).
- However, measurements were made in terms of a centimetre cube of a substance, giving ρ the units $\Omega \text{ cm}$.

Conductance(G)

- Is reciprocal of resistance(R)

$$G = \frac{1}{R} = \frac{1}{\rho \frac{l}{A}} = K \frac{A}{l}$$

where K (kappa): is reciprocal of specific resistance called as specific conductance or conductivity.

- It is measured in $\Omega^{-1} \text{ m}^{-1}$

- However, in SI system, the unit for conductance is '**Siemens**' and, given the **symbol 'S'**.
- Hence, the unit for conductivity or specific conductance will be $S\text{m}^{-1}$ ($1\text{S} = 1\Omega^{-1}$) or Scm^{-1} respectively.
 - remembered that $S\text{m}^{-1} = \mathbf{1/100} \text{Scm}^{-1}$.
- Further:

$$\rho = \frac{A \times R}{l}$$

$$\Rightarrow \frac{1}{\rho} = \frac{l}{A} \times \frac{1}{R}$$

$$\Rightarrow \kappa = \frac{l}{A} \times C$$

Here,
 $\frac{L}{A}$ *is cell constant*

Then, Specific conductance=conductance x cell constant.

Factors affecting conductance of electrolytes

- The conductance of an electrolytic solution depends mainly on
 - ✓ **the type of ions present,**
 - ✓ **speed with which the ions move in the solution and**
 - ✓ **their concentration.**

Factors affecting....

- **Temperature:**

- The conductivity of the electrolyte increases with the temperature increase.
 - Increase ions mobility

- **Concentration of the sample solution:**

- Inversely proportional to conductivity of the solution.

- **Number of ions in the sample solution:**

- The number of ions present in the solution is directly proportional to the conductance. (at diluted level)

- **Viscosity of the medium**

- Has inverse r/p

- **Pressure:** viscosity decrease by an increase pressure.

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Consequently, the equivalent conductance of the solution will

Molar Conductivity(Λ_m)

- The molar conductivity, Λ_m is the conductivity per unit molar concentration of a dissolved electrolyte.

$$\Lambda_m = k / C$$

Where

- C is the conc of the solution in Molar (mol dm^{-3})
- k (kappa – specific conductance) in $\Omega^{-1} \text{ cm}^{-1}$
- Since k (kappa – specific conductance) in $\Omega^{-1} \text{ cm}^{-1}$
 - then the following conversion is to be carried out
 - ✓ $\Lambda_m = 1000k / C$

09/25/2026 Thus the units of Λ_m are $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

Molar Conductivity increase with decreasing concentration for both weak and strong electrolytes.

➤ This is due to

➤ a decrease in **inter-ionic effects** for strong electrolytes and

➤ an increase in the **degree of dissociation** for weak electrolytes.

Molar Conductivity...

- At infinite dilution, **the ions are quite far apart**, the inter-ionic attractions are almost absent and each ion moves **completely independent of co-ions**.
- The molar conductivity then **approaches a limiting value at infinite dilution** and represents the conducting power of ***one mole of the electrolyte when it is completely split up into ions.***

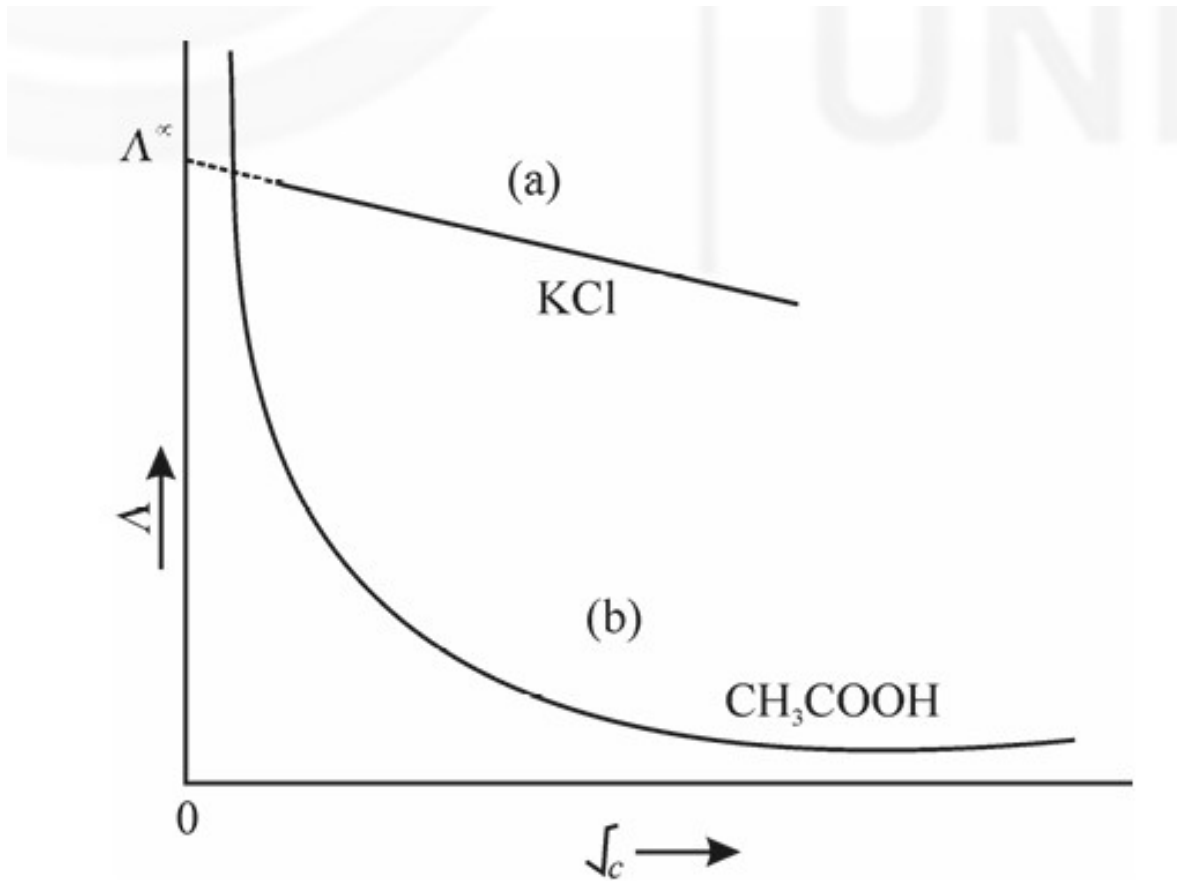


Fig. Variation of molar conductivity on dilution (a) for aqueous solution of potassium chloride (strong electrolyte) (b) acetic acid (weak electrolyte)

Molar Conductivity...

Conclusion

- In case of **weak electrolytes**, the increase in molar conductivity with dilution is mainly due to
 - (a) **an increase in the number of ions in the solution** (degree of ionization increases with dilution), and
 - (b) **smaller inter-ionic attractions** at higher dilutions (mostly useful for strong electrolytes).

Table: Molar conductivity of electrolytes in aqueous solutions at 298 K

$c/\text{mol dm}^{-3}$	KCl	NaCl	HCl	AgNO ₃	CH ₃ COOH	CH ₃ COONa
1.0	111.9	89.9	332.8	-	-	49.1
0.1	129.0	106.7	391.3	109.1	5.2	72.8
0.05	133.4	111.1	399.1	115.7	7.4	76.9
0.01	141.3	118.5	412.0	124.8	16.3	83.8
0.005	143.5	120.6	415.8	127.2	22.9	85.7
0.001	146.9	123.7	421.4	130.5	49.2	88.5
0.0005	147.8	124.5	422.7	131.4	67.7	89.2

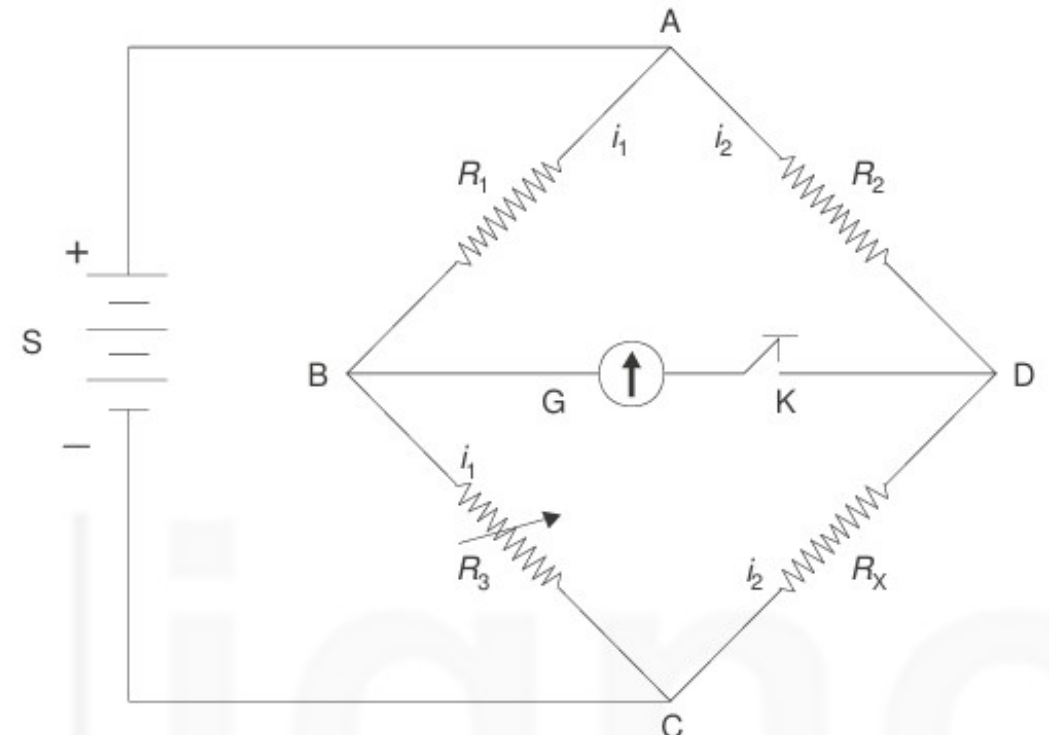
Measurement of electrolytic conductance

- The conductance of a solution can be determined by measuring the resistance offered by solution contained within the two electrodes of a conductivity cell.
- For measuring resistance, the **Wheatstone bridge principle** is employed.

The Wheatstone Bridge Principle

- A Wheatstone bridge is employed to measure the resistance of an electronic conductor.
- It works on the principle of obtaining balance between two arms with the help of a balance indicator (e.g. a galvanometer) at **the condition of potential being equal..**

- Let R_x be an unknown resistor, R_1 and R_2 two standard resistors, R_3 an adjustable resistor and G a galvanometer.
- The bridge is connected to a source of power S , a battery, and a tapping key K is placed in the path to control the connections.



Wheatstone bridge circuit

- To measure the resistance R_x , the tapping key K is held down momentarily and the bridge is balanced by adjusting R_3 to get no deflection in galvanometer under these conditions.
- In the bridge the total current is divided into two paths: i_1 through R_1 and R_3 , and i_2 through R_2 and R_x .
 - Hence, the potential at B (EB) must be equal to potential at D (ED).

$$EB=ED.....(1)$$

$$i_1 R_1 = i_2 R_2.....(2)$$

$$\text{Again } i_1 R_3 = i_2 R_x.....(3)$$

Devide eq 2 by eq 3

$$\frac{R_1}{R_3} = \frac{R_2}{R_x}$$

THEN

$$R_x = \frac{R_2 R_3}{R_1}$$

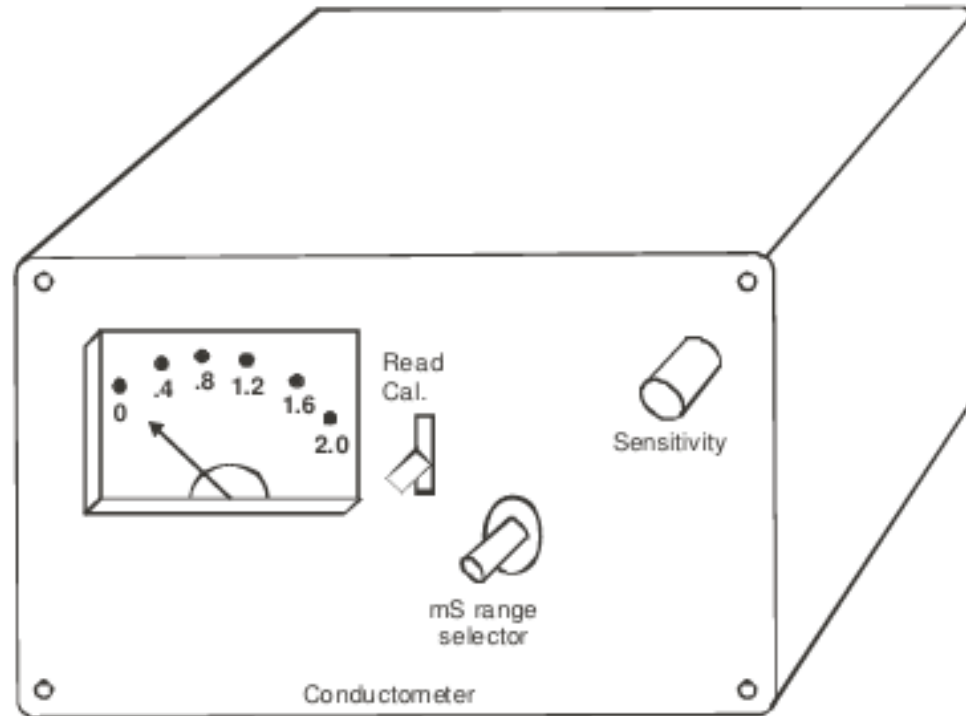
- Thus, we can calculate R_x as R_1 , R_2 and R_3 are all known.
- Conductance G , being the reciprocal of resistance will be

$$G = \frac{R_1}{R_2 R_3}$$

Conductometer

- From above discussion we can conclude that conductance is reciprocal of resistance and the resistance of a cell can be measured by placing it in an arm of a Wheatstone bridge.
- The inverse of the resistance gives the conductance and can be directly read on a conductivity measuring instrument, known as **“Conductometer”**.

- A typical conductometer, consists of an **ac source**, a **Wheatstone bridge** circuit, a **null detector or direct reading display** and a **conductivity cell**.
- Several inexpensive conductometers are commercially available



Conductometer

Applications

- **conductimetric titrations**

- Rate of change of conductance as a function of added titrant used to determine the end point.
 - Applied where visual or potentiometric methods fail to give results owing to considerable solubility or hydrolysis at the equivalence point.

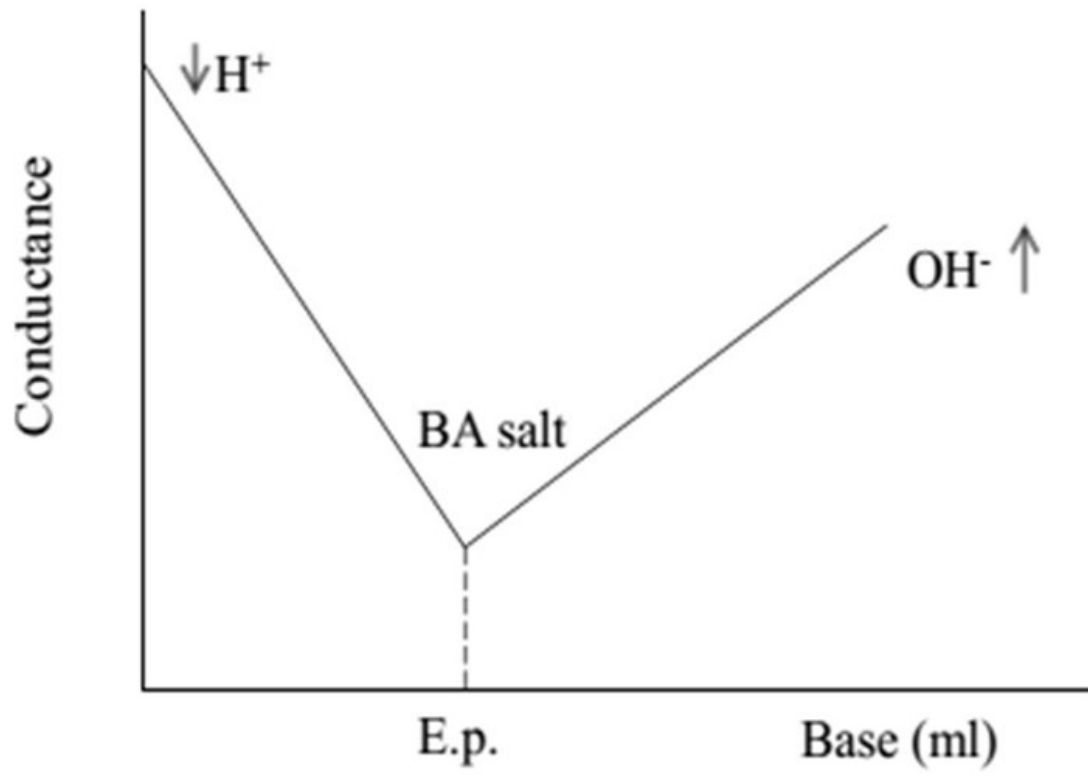
- Purity of distilled or de-ionized water is commonly checked by conductance measurement.

- Conductivity of pure water is about 5×10^{-8}

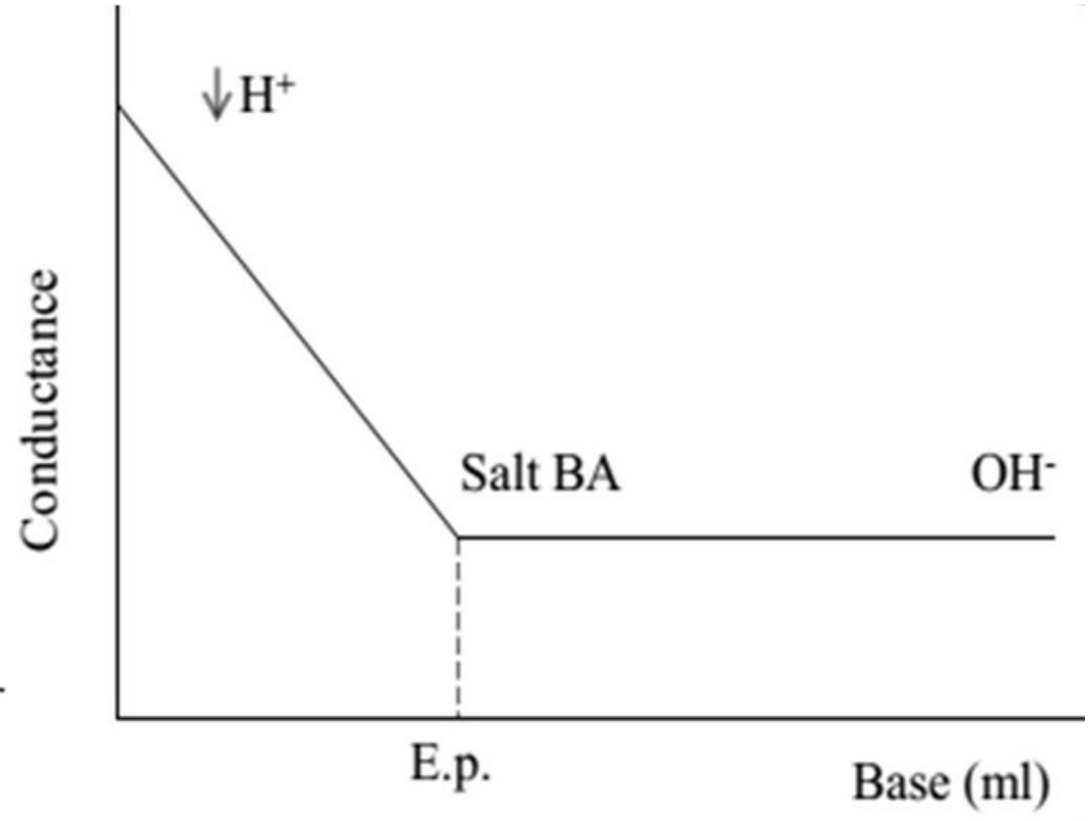
- Conductimetric monitoring of rivers & lakes water is used to control pollution.

- Conductimetric measurements are performed to determine salinity values

Titration of strong acid with a strong base

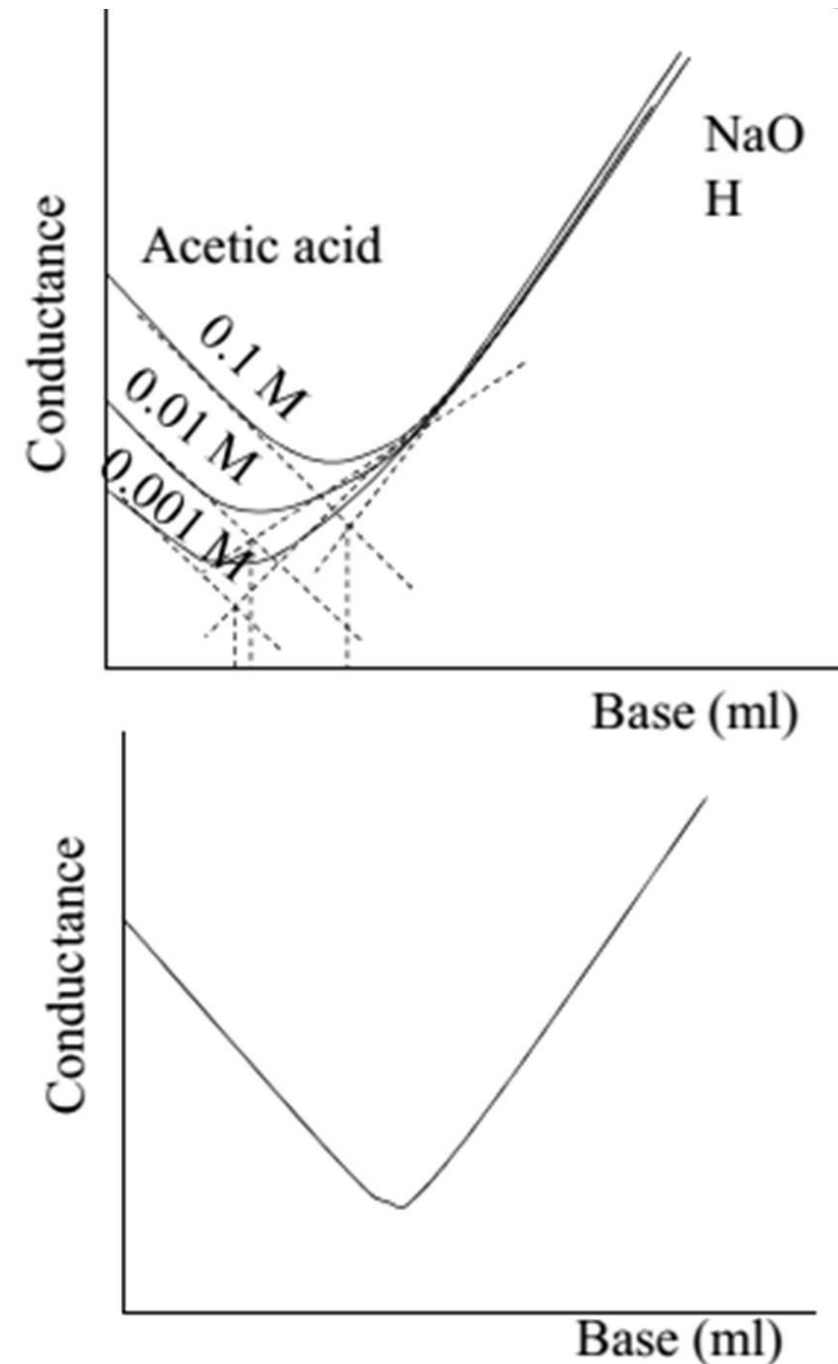


Titration of strong acid with a weak base



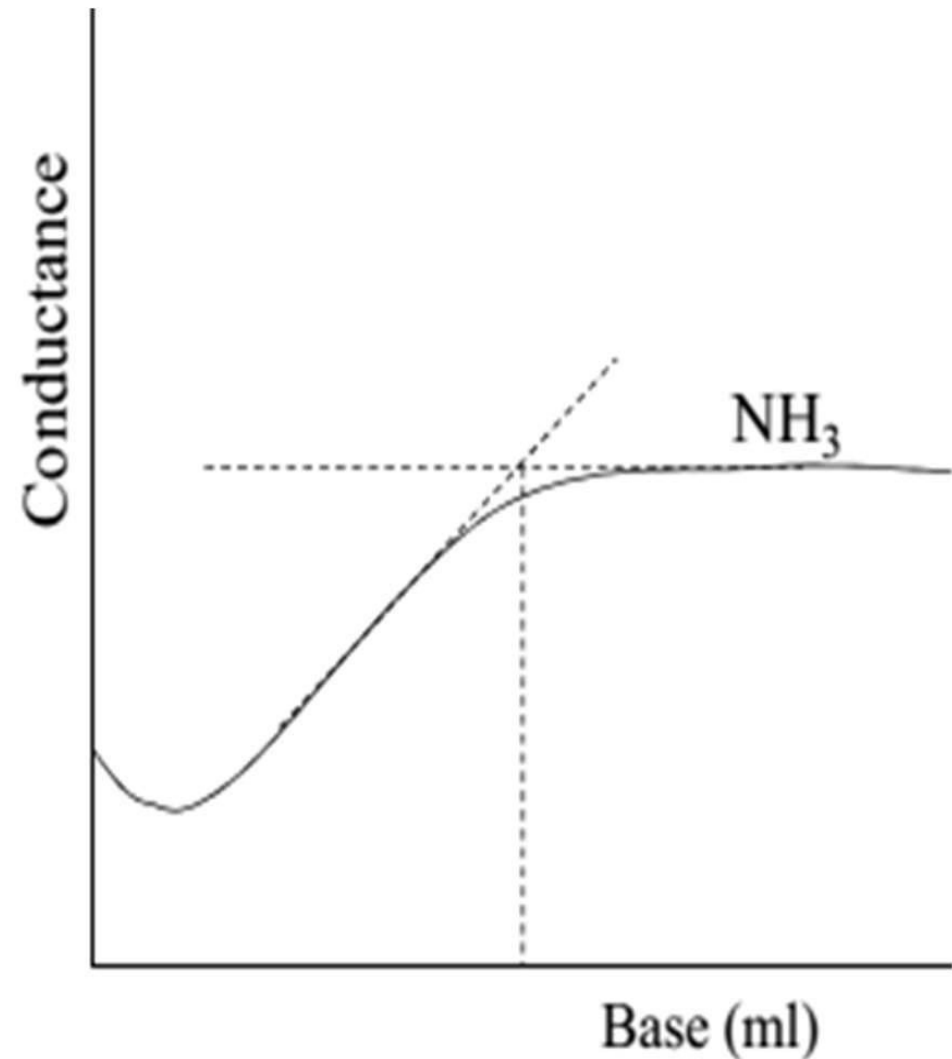
Titration of weak acid with a strong base

- The shape of the curve will depend upon the conc. & ionization of the acid.
- Thus the neutralization of the acetic acid ($K_a 1.8 \times 10^{-5}$) with NaOH solution, the salt which is formed during the first part of the titration tends to repress ionization of acetic acid still present so that its conductance decreases.
- The rising salt concentration will, however, tend to produce an increase in the conductance.
- In consequence of these opposing influences the titration curves may have minima, the position of which will depend upon the conc. and strength of the weak acid.
- As the titration proceeds, a somewhat indefinite break will occur at the



Titration of weak acids with weak bases

- Titration of a weak acid with a weak base can be readily carried out, & frequently it is preferable to use this procedure rather than use a strong base.
- The neutralization curve up to the equivalence point is similar to that obtained with NaOH since both sod. & amm. acetates are strong electrolytes, after the e.p. an excess of aqueous ammonia solution has little effect upon the conductance as its ionisation is depressed by the ammonia salt present in the solution.
- The advantages over the use of strong alkali are that the e.p. is easier to detect.



Titration of the mixture of a strong acid & a weak acid with a strong base

- Upon adding a strong base to a mixture of a strong acid and a weak acid (e.g. HCl & CH₃COOH), the conductance falls until the strong acid is neutralised, then rises as the weak acid is converted to its salt & finally rises more steeply as excess alkali is introduced.
- Note that:
 - Increasing ionisation of the weak acid results in a rounding-off at the first e.p.
 - Hydrolysis of the salt of the weak acid causes a rounding-off at the second e.p.
 - Extrapolation of the straight portions of the three branches leads to definite location of the end

